Peining Zhang (William)

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EDUCATION

Virginia Polytechnic Institute and State University Aug 2025 - Jun 2030 (expected)

- Ph.D. student in Industrial and Systems Engineering (ISE) United States
- Relevant coursework: Optimization for Machine Learning, Probability Theory

City University of Hong Kong Sep 2021 - Jun 2025

- Bachelor of Engineering in Intelligent Manufacturing (ITME) - cGPA: 3.88/4.30 (2/22) Hong Kong
- Dean's List (2021-2024); CityU Scholarship for Outstanding Students; Fung's ADSE Scholarship; ASM Pacific Technology Scholarship; SYE Student of the Year Award
- Relevant coursework: Machine Learning, Automation Technology, Mechanics

Peking University - Academic Exchange Programme Jul - Aug 2023

- Relevant coursework: Machine Learning Algorithms, Project Management Beijing
- Received the Special Fund for Non-local Outbound Exchange Students

TEACHING EXPERIENCE

Virginia Tech - Graduate Teaching Assistant (ISE 2214) Aug – Dec 2025

- Support course delivery through grading, student assistance, and coordination with the instructor on evaluation and course administration. Blacksburg, VA

RESEARCH/ PROJECT EXPERIENCE

Virginia Tech – Research Assistant May 2026 - Present

- Developed a Tensorized Fourier Neural Operator (TFNO3d) for transient 3D temperature-field prediction in laser powder bed fusion. Blacksburg, VA
- Achieved a test relative L2 error of 0.0757, MAE of 6.36 K, and RMSE of 12.84 K using Fourier modes (16, 16, 8), hidden width 24, projection width 48, and tensor rank 0.42.
- Developed a permutation-based quantum-inspired optimizer for LPBF scan-path planning using position and transition probability matrices and complete-route sampling without replacement.
- Coupled route generation with cached FDM thermal evaluation and complete-route fitness; designed elite soft-label and post-route local-credit updates and benchmarked the framework against classical GA and TSP-style baselines.

KTH Royal Institute of Technology - Visiting Researcher May - Aug 2024

- Built a MuJoCo and ROS 2 simulation for robotic EV-battery remanufacturing using dual 6-DoF ViperX arms and three RGB cameras. Sweden
- Designed the battery pack in SolidWorks, generated bimanual demonstrations, and adapted the mobile-ALOHA Action Chunking Transformer, achieving approximately 90% simulated task success and evaluating Diffusion Policy and VINN baselines.

City University of Hong Kong - Technical Assistant (Part Time) Mar - Dec 2023

- Co-authored a peer-reviewed literature review on digital twin technologies for civil infrastructure. Hong Kong
- Liu, C., Zhang, P., & Xu, X. (2023). Literature Review of Digital Twin Technologies for Civil Infrastructure. Journal of Infrastructure Intelligence and Resilience, 100050.

- Automated an Abaqus bridge model with Python and generated healthy and damaged structural-response data under randomized excitation.
- Developed CNN-LSTM and CNN-Transformer models with test accuracies of 99.66% and 97.26%; identified eight critical sensing nodes that retained 100% accuracy while reducing data by approximately 70%.

INTERNSHIP EXPERIENCE

Hebei King Ceramic Electronic Technology Co., Ltd. - Summer Intern May - Jun 2022

- Designed ceramic-substrate drawings in AutoCAD and 3D models in SolidWorks; gained hands-on exposure to production, chip packaging, connectivity testing, and C++. Hebei, China

LEADERSHIP EXPERIENCE/ OTHER ACHIEVEMENTS

BENG ITME Programme Committee, CityU - Student Representative Dec 2022 - Jun 2025

- Represented student needs in departmental planning and curriculum discussions and communicated student feedback to the department.

College Validation and Monitoring Committee (CVMC), CityU - Student Representative Sep 2022 - Jun 2025

- Reviewed undergraduate program proposals through committee meetings and paper circulation.

JPMorgan 2024 Asset and Wealth Management Challenge - Team Leader Mar 2024

- Led a four-member team in developing an asset-allocation proposal and delivering a 20-minute client presentation.

Mainland Student Association, CityU - Vice-president Sep 2022 - Sep 2023

- Led a 22-member team, established association regulations, and managed annual planning and weekly progress reviews.

ADDITIONAL INFORMATION

Languages: Native Mandarin, Proficient English, Elementary Cantonese

IT Skills: Python, PyTorch, MATLAB, C++, Qiskit, NumPy, SciPy, Git, Linux, CUDA, Abaqus, MuJoCo, ROS 2, SolidWorks, AutoCAD

Research Interests: Machine learning, optimization, digital twins, additive manufacturing, quantum-inspired computing, robotics